

3D Room Mapping LiDAR Scanner

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Executive Summary

This document details the process of creating a 3D Space Mapping LiDAR Scanner. This device is purposed to render a 3D map of an indoor environment. It is constructed by attaching a time of flight sensor to a rotating assembly allowing it to collect data with a 360 degree horizontal range, and a 90 degree vertical one. The scanner is operable remotely, by using SSH on a Raspberry Pi or by directly plugging in a computer. This project was incredibly intensive and educational in subjects such as data processing and visualization, mechanical engineering, and microcomputer systems.

Introduction

The 3D Space Mapping LiDAR Scanner is designed to explore the capabilities of 3D scanning. This technology is actively used in countless fields like robotics and autonomous driving. However, such systems can be incredibly expensive, with a single scanner costing thousands of dollars. This device, created with a budget of \$100, aims to recreate some of that functionality at a fraction of the cost.

Challenge summary

Throughout the development cycle, there were numerous key roadblocks that had to be overcome to produce the functioning scanner. These are detailed below:

Physical Ware:

- 3D Printer issues: Prints failed due to tangled filament and the printer required maintenance during the development cycle. We experienced issues with tolerances due to

printing orientation. Certain prints required careful utilization of supports to accommodate their complex features. There was a heavy learning curve associated with designing parts to balance assemblability, usability, and manufacturability.

- Mounting incompatible physicalware: 3D printed adapters were designed and fabricated to attach components that did not innately fit together. A small insert lies between the pulley gear and the servo arm, ensuring a tight fit. The rotating assembly is designed around the larger pulley, with an axle running down the middle. This axle provides space for slip ring wires and secures the assembly firmly to the pulley.

Hardware:

- Connecting wires: Slip ring wires did not have any connectors. To connect them, they needed to be soldered or spliced, neither of which were familiar techniques to the group members. This hindered our progress momentarily. As we decided to splice our wires we experienced a delay from needing to acquire heat shrink tubing and wire strippers.
- Power supply: After beginning integration, we discovered that our components drew more power than our Pi was able to provide. This required us to include an additional power supply connected directly to the other components. This added a delay in acquiring the power supply, as well as in rewiring the hardware.

Software:

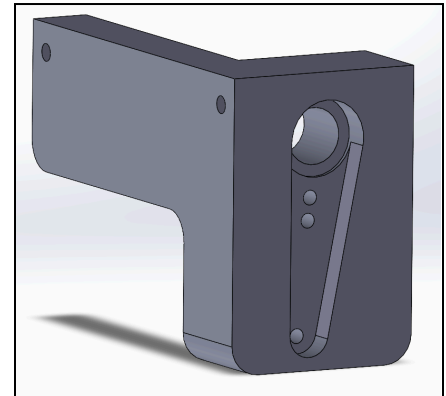
- Utilizing protocols: In setting up the software initially, it was challenging to get certain components to register. The most prominent example of this was the TF-Luna. This sensor is capable of operating in UART or in I2C. For our project, I2C was preferred, however, correctly setting up I2C was quite challenging, as it required careful software considerations in reading and writing to and from I2C addresses. This was also sort of a

hardware complication as to enable I2C on the TF-Luna, Pin 5 had to be grounded. This took a little bit of time to figure out, and required rewiring.

- **Visualization and Coordination:** A successful sensor needs to be able to accurately control where it wants to go as well as understand where it is. In this project, that means calculating stepper motor step sizes in relation to angles, servo motor angles, and cartesian mathematics for visualization.

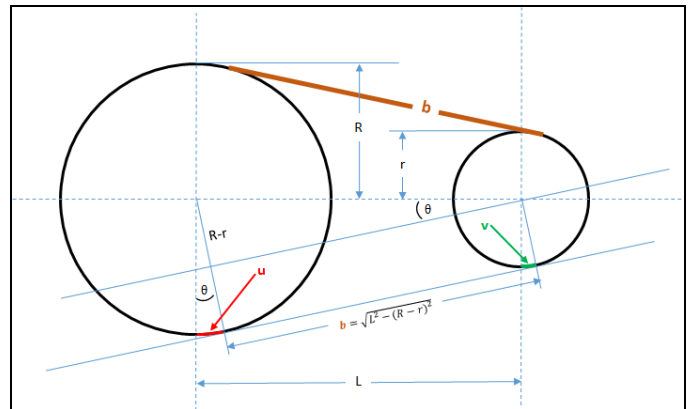
Problem solving & Contribution summary

The final project incorporates five self-designed 3D parts designed over the course of the project's lifecycle, with well over 15 total revisions. The main housing was reprinted five times alone. A notable example of the designwork involved in this project was the sensor mount, which was the final part created. While small, this part introduced unique challenges in design, fabrication, and assembly. The



positioning between the sensor and servo had to be precise to prevent discrepancies between reality and software, which would result in inaccurate measurements. Solving this problem meant optimizing tolerances between PLA and nylon, three different types of screw with the added constraint of being printable. On one print, tolerances were too tight due to the printing orientation chosen during slicing. On one another, all parts fit together correctly, yet a slight design flaw rendered the system impossible to assemble. The resulting part is functional and robust, highlighting just one example of the types of problems solved in the physical design of the LiDAR scanner.

Another consideration was the design of the pulley system. This required significant planning and mechanical engineering to successfully implement. The intentionality behind the horizontal pulley positioning can be visualized in the figure to the right. In addition to this, the correct relative heights between the stepper arm and the rotating assembly needed to be determined, keeping in mind adequate space for the slip ring. Attaching the rotating assembly was an initial challenge, but was solved relatively quickly

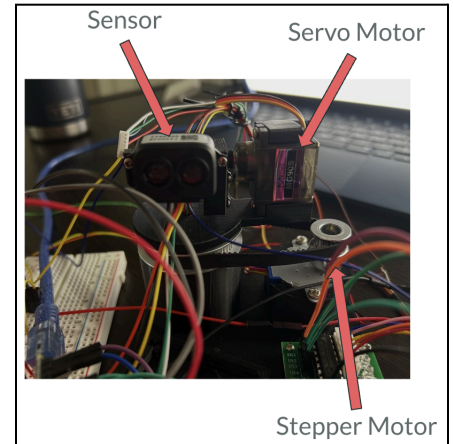


System Design

The physical ware consists of several key components. A fully 3D printed base holds the Pi, ULN2003 stepper motor driver, stepper motor, and supports a rotating assembly with a slip ring. Resting atop of the base is a 360-degree rotating assembly, driven by a GT-2 pulley and 28BYJ-48 stepper motor. This base supports the sensor assembly which is discussed in the hardware section below.

The hardware can be thought of in terms of stationary components and rotating ones. On the stationary side, a computer or Raspberry Pi 4 runs control software. This allows for easy interfacing and optional mobility via SSH. To drive other components, an Arduino Nano is used. This is preferred over the Pi for its real time operating system (RTOS) which allows it to better coordinate all of the different components without experiencing unnecessary context switching.

On the rotating assembly, an MG90s servo is used to drive the vertical rotation of A TF-Luna. The MG90s is preferable over other options due to its rigid metal construction and low power consumption. Similarly, the TF-Luna's low power draw, affordable price, and 8m max range made it ideal for this project. The wiring between stationary components and rotating ones is managed with a slip ring, which lies right below the rotating assembly. To aid in the development of the hardware, a comprehensive wiring chart was created:



TF Luna	Slip Ring	Stationary
1. 5V (Black)	Brown	Nano 5V
4. Ground (red)	Black	GND
2. SDA (White)	Yellow	Nano A4
3. SCL (Green)	Green	Nano A5
5. I2C Enable (Yellow)	Black	GND

*Important to note that the TF Luna has 5V and GND wire colors FLIPPED

MG90S Servo	Slip Ring	Stationary
5V (Red)	Red	PSU
GND (Brown)	Black	GND
PWM (Orange)	Orange	Nano D3

ULN2003 Driver Board	Stationary
5V (+)	PSU
GND (-)	GND

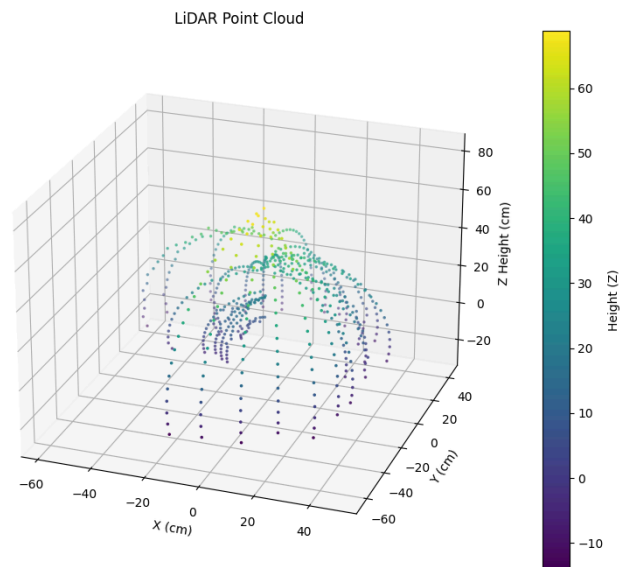
IN1	Nano D5
IN2	Nano D6
IN3	Nano D7
IN4	Nano D8
Output pin array	28BYJ-48 Stepper

Alternatively, the software design needed to allow both precise operation of components, visualization, and user input. As such, the resulting software is composed of three files. The first of which is the firmware for the Nano. This software polls for a command over its serial bus. This control command is sent via the computer or Pi through the means of a Python script. This allows the user to choose between a low resolution quick scan, and a high resolution slower one. When a signal is received, the nano initiates a scan without interruption using calibrated parameters. This resulting data is communicated back via the serial bus to the sending device, which can then run a visualizer script written in Python. This script is written using common mathematics and visualization libraries: Pandas and Matplotlib.

Outcomes

We successfully built a prototype LiDAR scanner that maps its surroundings and produces a 3D visualization. Visualization shows measurement data, which can be roughly used for measuring. The scanner can be operated with a decent amount of mobility due to the onboard Pi via SSH. While the final product is able to deliver on its fundamental goal, its level of polish and featureset could be improved upon. For instance, the LiDAR is limited by the speed of the stepper and servo motors as well as by its immobility, upgrades to the assembly mechanics and to the stepper motor could greatly increase the speed of the scanner. The addition of a homing

feature could also increase the accuracy of the scanner, importantly allowing for multiple scans to be layered on top of each other with consistent positioning. The wiring and lack of attachment of the Nano is a point that could be improved upon, but was deemed ultimately less important than ensuring functionality of the scanner. Given more time, managing these wires better by shortening them, tying them, and placing a shroud over the entire assembly would greatly improve it. The software also has room for expansive improvements. Notable ones include consolidating the different python scripts into a unified program, adding more control features (adjustable spin rate, speed, min and max rotation), expanding visualization and processing through noise filtration algorithms, measurement capabilities, and mesh creation, could improve its overall utility.



Results of an early fast scan.

References

- TF-Luna LiDAR Datasheet:
<https://en.benewake.com/uploadfiles/2024/04/20240426135921367.pdf>
- TF-Luna LiDAR Guide:
<https://makersportal.com/blog/distance-detection-with-the-tf-luna-lidar-and-raspberry-pi>
- Stepper Motor Datasheet:
<https://www.mouser.com/datasheet/2/758/stepd-01-data-sheet-1143075.pdf?srltid=AfmBOoo1YpLiJBofaRnlbLaHaxJNy-KeU7Joi4tX3U540zWgguQCvvtF>
- Pulley calculation & diagram:
<https://www.vcalc.com/equation/?uuid=788be105-ed67-11e3-b7aa-bc764e2038f2>